

Jeff Shen

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Education

H.B.Sc., Statistics, Astronomy & Astrophysics, Mathematics
University of Toronto

2018-2022

Technical Skills

Data: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Stan, Seaborn, SQL, NoSQL, R, Excel

Web: HTML5, CSS3, JavaScript, React (Native), Redux, Node, Express.js, Flask

Other: Rust, Docker, shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, SymPy, Astropy, Selenium, NLTK, $\text{\LaTeX}$, regular expressions

Research Experience

- 2020 Aug. - Present **Undergraduate Research Course**
Canadian Institute for Theoretical Astrophysics
Supervised by [Prof. Gwendolyn Eadie](#) and [Prof. Norman Murray](#). Study of mass distribution of the Milky Way in a Bayesian framework.
- 2020 May - 2020 Aug. **Summer Undergraduate Research Program**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto
Supervised by [Dr. Allison Man](#). Study of molecular gas in a strongly-lensed protocluster system at $z \approx 2.9$.

Honours and Awards

- 2020 **Summer Undergraduate Research Program Award (\$9,500 CAD)**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Independent Coursework

- 2020 **Practical Time Series Analysis**, The State University of New York (Coursera)
Stationarity, differencing, ACF, PACF, autocovariance. MA processes, AR processes, AR-MA duality. Backshift operator, difference equations, Yule-Walker matrix equations. AIC, SSE, model selection. ARMA and ARIMA models. Seasonality, SARIMA. Exponential smoothing, forecasting.
- 2019 **Data-Driven Astronomy**, The University of Sydney (Coursera)
Pulsars, black holes, exoplanets, stellar evolution, distances, redshift, galaxy classification. Image stacking, algorithm analysis, crossmatching, databases, regression, classification, machine learning. Implementation in Python with Astropy; SQL.
- 2019 **Intro to Deep Learning with PyTorch**, Udacity
Mathematics of neural networks. Data preparation, regularization techniques. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.
- 2018 **Intro to Machine Learning**, Udacity
Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, feature selection, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with scikit-learn.

Outreach

- 2019 - Present **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy.
Episodes 177 - present.

Volunteering

- 2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.

Relevant Coursework

2020 Fall	Real Analysis
2020 Fall	Regression Analysis
2020 Fall	Practical Astronomy
2020 Winter	Galaxies and Cosmology
2020 Winter	Practical Physics I
2020 Winter	Electricity and Magnetism
2020 Winter	Thermodynamics and Statistical Mechanics
2019 - 2020	Probability and Statistics I, II
2019 Fall	Stars and Planets
2019 Fall	Ordinary Differential Equations
2019 Summer	Multivariable Calculus
2019 Winter	Computer Science
2018 - 2019	Calculus
2018 - 2019	Linear Algebra I, II
2018 Fall	Statistical Reasoning and Data Science

Interests

Languages: English (native), French, Chinese, Norwegian (basic)

Other: *nix customization, computer graphics, 3D modelling, running, baking, reading, guitar